BABS1202 Practical test information

This is an in-class test where you will perform laboratory techniques that you have practiced multiple times in the laboratory classes, to assess these three competencies:

1. Presumptively identify bacterial samples through Gram staining and light microscopy.
2. Use aseptic culturing techniques to safely isolate and culture microorganisms.
3. Quantify the number of microorganisms in a sample using serial dilution and viable plate count techniques.

The laboratory techniques you will perform are listed below:

1. Perform a Gram stain on a cell culture (Competency 1)
2. Independently focus a field of view using the light microscope and identify the Gram status of the cells (Competency 1)
3. Use the streak plate method to subculture bacteria and obtain single colonies (Competency 2)
4. Subculture bacteria by performing a broth culture inoculation (Competency 2)
5. Maintain aseptic technique when transferring a sterile liquid from one container to another (Competency 2).
6. Perform 10-fold serial dilutions and use the spread plate method to obtain a viable plate count i.e. a countable number of colonies in the predicated range of the dilution (Competency 3)
7. Perform serial dilutions in a 96 well plates (Competency 3)

Due Date and weighting

- Scheduled Week: Your scheduled laboratory class in Week 9
- Duration: 90 minutes
- Location: Your scheduled laboratory class
- This task is worth 20% of your final grade in the course

Criteria

Task 1: Perform a Gram Stain on a Cell Culture (Competency 1)
No marks: Uneven distribution of bacterial cells AND incorrect Gram-status for stained sample.
One mark: Uneven distribution of bacterial cells OR incorrect Gram-status for stained sample.
Two marks: Even distribution of bacterial cells AND correct Gram-status for stained sample.

Task 2: Independently Focus a Field of View Using the Light Microscope and Identify the Gram Status of the Cells (Competency 1)
No marks: Unable to independently focus on a field of view AND incorrectly identify Gram-status and shape of visualised sample.
One mark: Unable to independently focus on a field of view (but demonstrator assistance allows for identification of the Gram-status) OR incorrectly identify Gram-status and shape of visualised sample.
Two marks: Independently focus on a field of view AND correctly identify Gram-status and shape of visualised sample.

Note: if the Gram stain and even distribution of cells is not performed in Task 1, the demonstrator can provide a sample for Task 2.

Task 3: Use the Streak Plate Method to Subculture Bacteria and Obtain Single Colonies (Competency 2)
Zero marks: Absence of any dilution in microbial growth via streaking.
One mark: Successful isolation of single colonies following streaking from the primary inoculum.

Task 4: Maintain Aseptic Technique When Transferring a Sterile Liquid from One Container to Another (Competency 2)
Zero marks: Presence of turbidity (contamination) in BOTH bottles following incubation.
One mark: Presence of turbidity in EITHER bottle following incubation.
Two marks: BOTH bottles remain sterile following broth transfer and incubation.

Task 5: Subculture Bacteria by Performing a Broth Culture Inoculation (Competency 2)
Zero marks: Absence of turbidity in inoculated bottle following incubation.
One mark: Presence of turbidity in inoculated bottle following incubation.

Task 6: Perform 10-Fold Serial Dilutions and Use the Spread Plate Method to Obtain a Viable Plate Count (Competency 3)
Zero marks: Number of colonies on final dilution plate differs by more than an order of magnitude compared to predicted number.
One mark: Number of colonies on final dilution plate within same order of magnitude as predicted number.

Task 7: Perform Serial Dilutions in a 96-Well Plate (Competency 3)
Zero marks: Incorrect 1:10 serial dilution of food dye across 8 wells.
One mark: Correct 1:10 serial dilution of food dye across 8 wells.

Total marks available = 10 marks

Generative AI

Planning assistance: You may use generative AI to help you plan for the Practical test, such as by summarising the methods into a format you can follow.

Please note that the outputs from these tools are not always accurate, appropriate, nor properly referenced. You should ensure that you have moderated and critically evaluated the outputs from generative AI tools such as ChatGPT with those provided in your Class Notebook.

If the outputs of generative AI such as ChatGPT form part of your submission in class it will be regarded as serious academic misconduct and subject to the standard penalties, which may include 00FL, suspension and exclusion.

Course Learning Outcomes assessed

CLO2 : Demonstrate a knowledge of the skills, techniques, experimental design and safe work practices of a bioscience laboratory, including fundamental microbiology, recombinant DNA and bioinformatics techniques.
CLO4 : Demonstrate quantitative skills applicable to the biosciences, including serial dilutions, metric conversions, biochemical calculations data collection and organisation, graphing, and experimental design.

Instructions

You need to arrive on time for the Practical test and have the appropriate PPE for the laboratory. You will be provided with all the equipment you need to perform the techniques assessed.

Task 1: Perform a Gram Stain on a Cell Culture (Competency 1)
You will be given a liquid bacterial culture. You need to perform a Gram stain with an even distribution of bacterial cells that is stained for identification of Gram status.
Feedback: Your demonstrator will view your sample and assess using a rubric.

Task 2: Independently Focus a Field of View Using the Light Microscope and Identify the Gram Status of the Cells (Competency 1)
You need to use the light microscope to focus a field of view that allows you to identify the Gram status of the cells. You should use your sample from Task 1, but can be provided a sample if you did not successfully complete Task 1. You will record your identification on the answer sheet.
Feedback: Your demonstrator will view your sample and mark your answer sheet assess using a rubric.

Task 3: Use the Streak Plate Method to Subculture Bacteria and Obtain Single Colonies (Competency 2)
You will be given a liquid bacterial culture and use this to subculture the bacteria using the streak plate method.
Feedback: Your demonstrator will view your plate following incubation and assess using a rubric. You will be able to view your plate in Week 10.

Task 4: Maintain Aseptic Technique When Transferring a Sterile Liquid from One Container to Another (Competency 2)
You will be given two containers of sterile broth. You need to transfer a given volume from one to the other using a pipette.
Feedback: Your demonstrator will view your samples for turbidity the following incubation and assess using a rubric. You will be able to view your containers in Week 10.

Task 5: Subculture Bacteria by Performing a Broth Culture Inoculation (Competency 2)
You will be given a liquid bacterial culture and use this to inoculate a broth culture.
Feedback: Your demonstrator will view your sample following incubation and assess using a rubric. You will be able to view your plate in Week 10.

Task 6: Perform 10-Fold Serial Dilutions and Use the Spread Plate Method to Obtain a Viable Plate Count (Competency 3)
You will be given a bacterial broth culture, dilutant, and tubes with which to perform the serial dilution, and will spread the final dilution on an agar plate.
Feedback: Your demonstrator will view your plate following incubation and assess using a rubric. You will be able to view your plate in Week 10.

Task 7: Perform Serial Dilutions in a 96-Well Plate (Competency 3)
You will be given a dye, dilutant, and microtitre plate with which to perform the serial dilution.
Feedback: Your demonstrator will view your plate on the day of the test and assess using a rubric.